

Heishigan Pathmaraj

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PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Developer with expertise in AI-driven system development and multi-agent frameworks. Holds a First-Class BSc in Computer Science and an MSc in AI for Health. Experienced in building SaaS applications and AI-powered solutions, combining rigorous AI research with production-grade software engineering. Skilled in collaborating with both technical and non-technical stakeholders to deliver impactful AI services.

EDUCATION

Master of Science in AI for Health

Sep 2024 – Jun 2026

Stockholm University | Stockholm, Sweden

Bachelor of Science in Computer Science

Oct 2019 – Sep 2022

Staffordshire University | Stoke-on-Trent, UK | First Class Honours

PROJECTS

Cursiva.se: Autonomous AI Job Application Pipeline

Jul 2026

Shipped SaaS App | Python, FastAPI, LangGraph, Next.js, PostgreSQL

- Architected an end-to-end multi-agent pipeline using LangGraph and FastAPI, acting as an autonomous executive assistant to tailor CVs and dynamically compile native LaTeX PDFs
- Built a production-ready full-stack system featuring a Next.js 14 frontend, Supabase PostgreSQL data layer, and Clerk JWT authentication, using Server-Sent Events (SSE) to stream live agent progress

TriageSim: Multi-Agent Framework for LLM Clinical Triage Evaluation *Jan – Jun 2026*

MSc Thesis, Stockholm University | Python, LLMs, RAG, FAISS, NLP, EHRs

- Built a 3-agent system (Patient, Nurse, Evaluator) to simulate LLM-driven clinical triage of elderly fall patients across N=1,000 encounters using de-identified MIMIC-IV EHR records
- Reduced dangerous under-triage (Lethal Miss Rate) from 28.4
- Ran an OSCE-style human pilot study; all perceptual dimensions rated above 4.0/5.0 across realism, clinical reasoning quality, and perceived utility

Virtual Diabetes Triage API

Oct 2025

Stockholm University | Python, FastAPI, Docker, GitHub Actions, Scikit-learn, SQL

- Deployed a containerised ML microservice scoring short-term diabetes progression risk via a versioned REST API, iterating from LinearRegression to RandomForestRegressor; final model achieved 73.9
- Automated the full delivery pipeline with Docker containerisation, GitHub Actions CI/CD, and a SQL-backed data layer; API response under 100 ms with 7/7 tests passing

Vaccine Misinformation Detector

May 2025

Stockholm University | Python, BERT, NLP, OCR, RAG, Hugging Face

- Fine-tuned BERT for binary vaccine misinformation classification, achieving 94
- Architected a RAG pipeline for evidence-grounded, citation-backed rebuttal generation

Prototype of an Affective Video Game

Aug 2022

BSc Final-Year Project | Python, Arduino, Facial Emotion Recognition

- Built a horror game that adapts gameplay in real-time to the player’s biometric state via facial emotion recognition and an Arduino heartbeat sensor

WORK EXPERIENCE

Loading Manager & Delivery Rider

Oct 2024 – Present

Velove Bikes AB | Stockholm

- Manage daily hub operations in a high-volume, commercially driven logistics environment, maintaining performance standards and ensuring consistent on-time delivery

TECHNICAL SKILLS

Programming:	Python, SQL, JavaScript (ReactJS), Java, C#, C++, HTML/CSS, Next.js
AI & Data:	LLMs, Multi-Agent Systems, RAG, NLP, BERT, TensorFlow, FAISS, Scikit-learn, pandas, LangGraph
Infrastructure:	Docker, GitHub Actions, FastAPI, CI/CD Pipelines, MLOps, Git, Linux
Game Dev:	Unreal Engine 5 (C++ & Blueprint), Arduino, Biofeedback Systems
Languages:	English (Fluent), Swedish (B2, SFI and SVA Grund completed)
Other:	Swedish Driver’s Licence (Körkort AM/B)